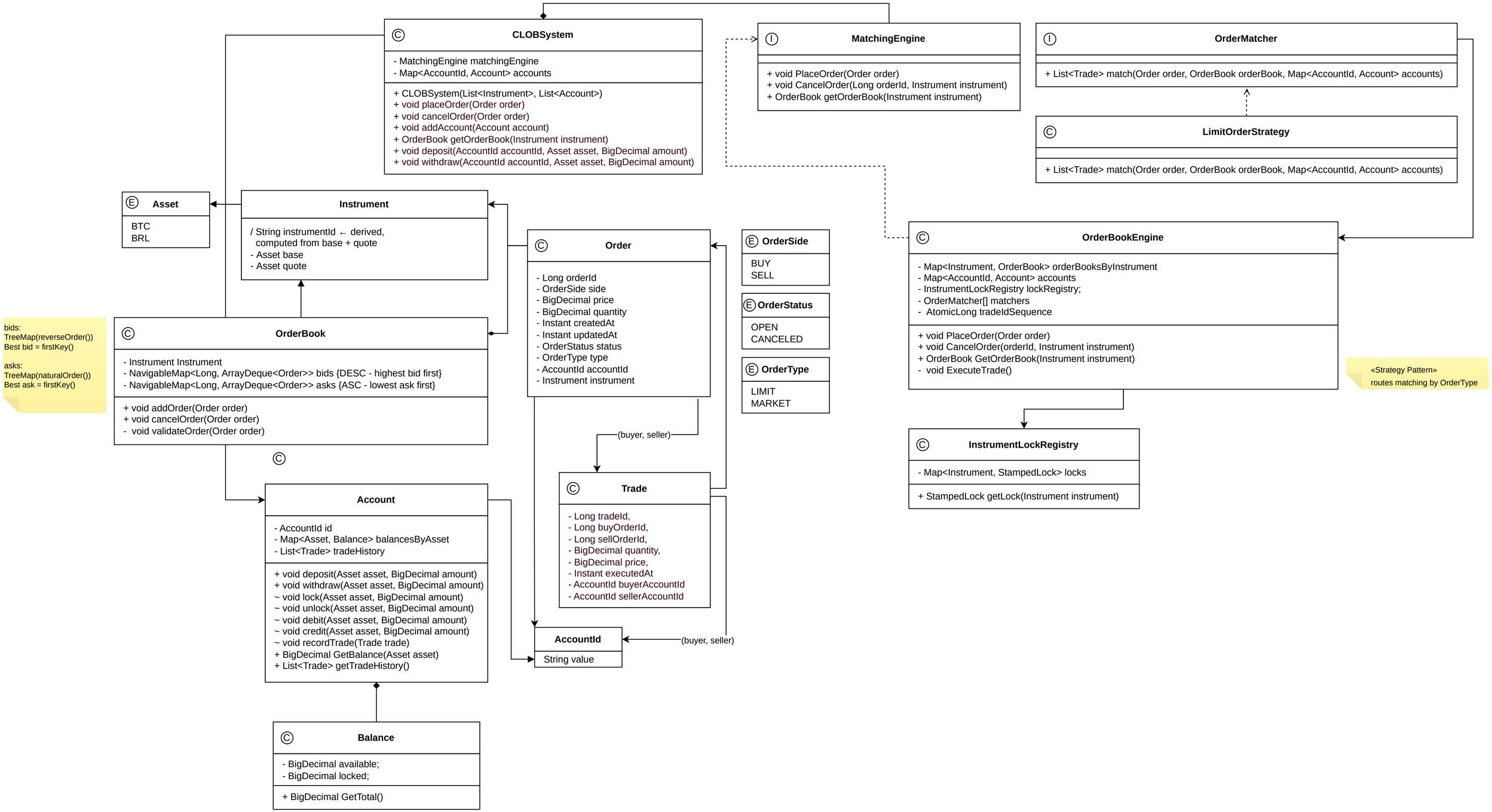
CLOBSystem
- MatchingEngine matchingEngine
- Map<AccountId, Account> accounts
+ CLOBSystem(List<Instrument>, List<Account>)
+ void placeOrder(Order order)
+ void cancelOrder(Order order)
+ void addAccount(Account account)
+ OrderBook getOrderBook(Instrument instrument)
+ void deposit(AccountId accountId, Asset asset, BigDecimal amount)
+ void withdraw(AccountId accountId, Asset asset, BigDecimal amount)
MatchingEngine
+ void PlaceOrder(Order order)
+ void CancelOrder(Long orderId, Instrument instrument)
+ OrderBook getOrderBook(Instrument instrument)
OrderMatcher
+ List<Trade> match(Order order, OrderBook orderBook, Map<AccountId, Account> accounts)
LimitOrderStrategy
+ List<Trade> match(Order order, OrderBook orderBook, Map<AccountId, Account> accounts)
Asset
BTC
BRL
Instrument
/ String instrumentId ← derived,
computed from base + quote
- Asset base
- Asset quote
Order
- Long orderId
- OrderSide side
- BigDecimal price
- BigDecimal quantity
- Instant createdAt
- Instant updatedAt
- OrderStatus status
- OrderType type
- AccountId accountId
- Instrument instrument
OrderSide
BUY
SELL
OrderStatus
OPEN
CANCELED
OrderType
LIMIT
MARKET
OrderBookEngine
- Map<Instrument, OrderBook> orderBooksByInstrument
- Map<AccountId, Account> accounts
- InstrumentLockRegistry lockRegistry;
- OrderMatcher[] matchers
- AtomicLong tradeIdSequence
+ void PlaceOrder(Order order)
+ void CancelOrder(orderId, Instrument instrument)
+ OrderBook GetOrderBook(Instrument instrument)
- void ExecuteTrade()
«Strategy Pattern»
routes matching by OrderType
bids:
TreeMap(reverseOrder())
Best bid = firstKey()
asks:
TreeMap(naturalOrder())
Best ask = firstKey()
OrderBook
- Instrument Instrument
- NavigableMap<Long, ArrayDeque<Order>> bids {DESC - highest bid first}
- NavigableMap<Long, ArrayDeque<Order>> asks {ASC - lowest ask first}
+ void addOrder(Order order)
+ void cancelOrder(Order order)
- void validateOrder(Order order)
(buyer, seller)
InstrumentLockRegistry
- Map<Instrument, StampedLock> locks
+ StampedLock getLock(Instrument instrument)
Trade
- Long tradeId,
- Long buyOrderId,
- Long sellOrderId,
- BigDecimal quantity,
- BigDecimal price,
- Instant executedAt
- AccountId buyerAccountId
- AccountId sellerAccountId
Account
- AccountId id
- Map<Asset, Balance> balancesByAsset
- List<Trade> tradeHistory
+ void deposit(Asset asset, BigDecimal amount)
+ void withdraw(Asset asset, BigDecimal amount)
~ void lock(Asset asset, BigDecimal amount)
~ void unlock(Asset asset, BigDecimal amount)
~ void debit(Asset asset, BigDecimal amount)
~ void credit(Asset asset, BigDecimal amount)
~ void recordTrade(Trade trade)
+ BigDecimal GetBalance(Asset asset)
+ List<Trade> getTradeHistory()
AccountId
String value
(buyer, seller)
Balance
- BigDecimal available;
- BigDecimal locked;
+ BigDecimal GetTotal()